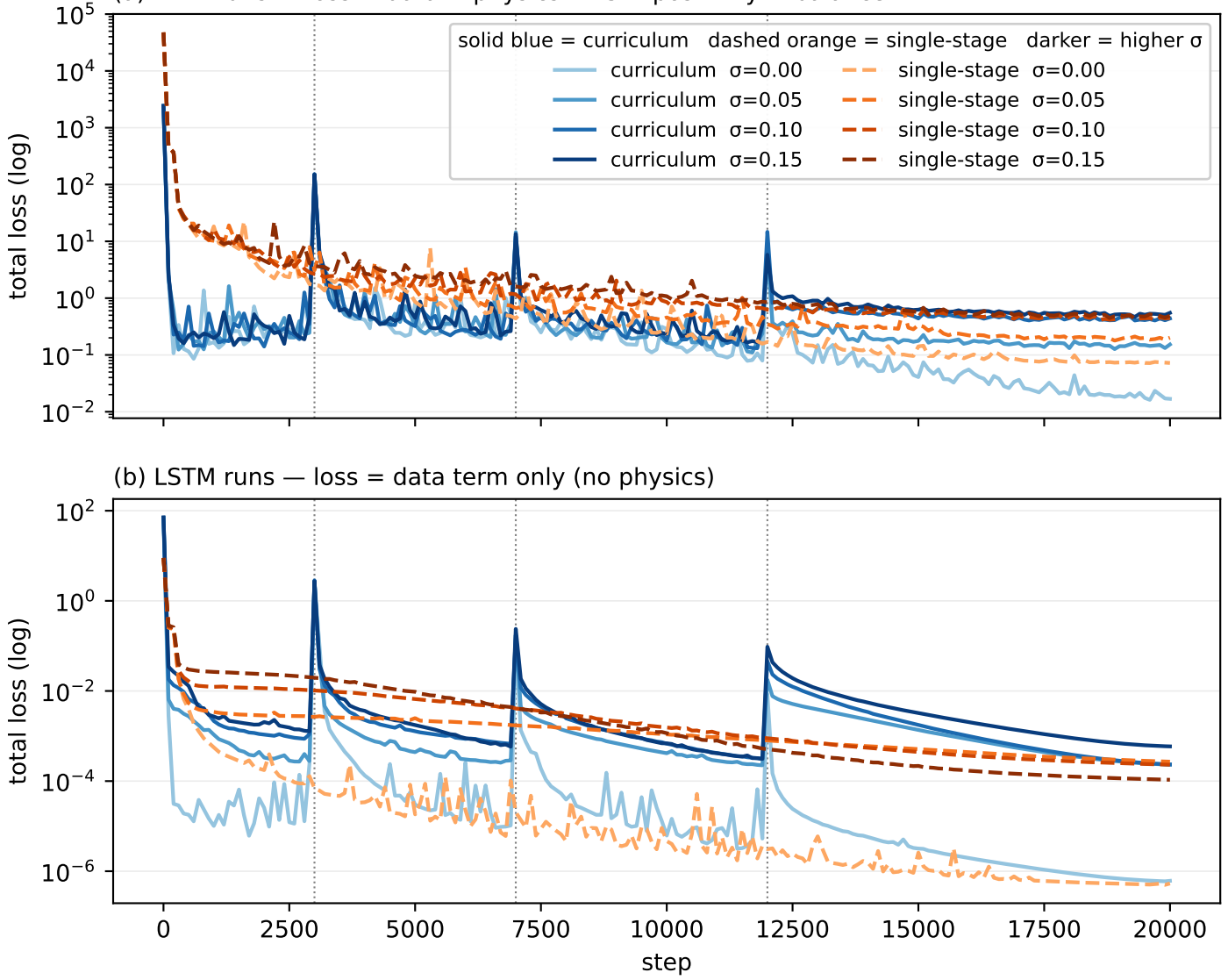
(a) PINN runs — loss = data + physics + IC + positivity + balance
total loss (log)
solid blue = curriculum dashed orange = single-stage darker = higher σ
curriculum σ=0.00
curriculum σ=0.05
curriculum σ=0.10
curriculum σ=0.15
single-stage σ=0.00
single-stage σ=0.05
single-stage σ=0.10
single-stage σ=0.15
(b) LSTM runs — loss = data term only (no physics)
total loss (log)
0
2500
5000
7500
10000
12500
15000
17500
20000
step